

Spring Boot - Student Notes

Introduction, Features, Architecture, Starters, Embedded Server and Interview Points

1. What is Spring Boot?

Spring Boot is built on top of the Spring Framework. It simplifies the development of standalone, production-ready Spring applications.

Spring Boot = Spring Framework + Auto Configuration + Starter Dependencies + Embedded Server + Production-ready features

2. Why Do We Need Spring Boot?

Traditional Spring applications can require significant manual configuration. Spring Boot reduces this work so developers can focus more on business logic.

- Reduces manual Spring configuration
- Simplifies dependency management
- Provides an embedded web server
- Makes database and web integration easier
- Allows applications to run directly as executable JAR files

3. Key Features of Spring Boot

Auto Configuration

Spring Boot automatically configures many application components based on the dependencies available in the project and the application's configuration.

```
org.springframework.boot  
spring-boot-starter-web
```

Starter Dependencies

Starter dependencies are predefined dependency groups. Instead of adding many related libraries individually, we add a suitable Spring Boot starter.

Starter	Purpose
spring-boot-starter-web	REST APIs and Spring MVC web applications
spring-boot-starter-data-jpa	Database access using JPA and Hibernate
spring-boot-starter-security	Application authentication and security
spring-boot-starter-validation	Request and data validation
spring-boot-starter-test	Unit and integration testing
spring-boot-starter-actuator	Health, metrics and monitoring

4. Embedded Server

Spring Boot web applications can include the web server inside the application. The common default for Spring MVC applications is **Tomcat**. Jetty or Undertow can also be used.

```
mvn spring-boot:run

java -jar employee-service.jar
```

Because the server is embedded, we normally do not need to install and deploy the application manually to a separate Tomcat installation.

5. Standalone Application

A Spring Boot application can be started directly from its main method.

```
@SpringBootApplication
public class EmployeeApplication {

    public static void main(String[] args) {
        SpringApplication.run(EmployeeApplication.class, args);
    }
}
```

6. @SpringBootApplication

This is the main convenience annotation commonly placed on the application's startup class. It enables three important capabilities:

- **@Configuration** - allows the class to provide Spring configuration.
- **@EnableAutoConfiguration** - enables Spring Boot's auto-configuration mechanism.
- **@ComponentScan** - scans application packages for Spring-managed components.

Common scanned components include `@Component`, `@Service`, `@Repository`, `@Controller` and `@RestController`.

7. How Spring Boot Works

```
Developer
|
Application Code
|
Spring Boot
|
Auto Configuration
|
Spring Framework
|
Embedded Server
|
JVM / Operating System
```

Important: Spring Boot does not replace the Spring Framework. It uses Spring Framework internally and makes its configuration and application development easier.

8. Typical REST API Architecture

```
Client / Postman
|
Controller
|
```

```

    Service
    |
Repository
    |
JPA / Hibernate
    |
Database

```

Controller: Handles HTTP requests and responses.

Service: Contains business logic.

Repository: Communicates with the persistence layer/database.

9. Spring Boot Architecture

```

Application Code
Controller / Service / DTO
    |
Spring Boot Auto Configuration
    |
Spring Framework
Core | MVC | Data | Security
    |
Embedded Server
Tomcat / Jetty / Undertow
    |
JVM / Operating System

```

10. Convention Over Configuration

Spring Boot provides sensible defaults so developers do not have to configure everything manually. These defaults can still be overridden when required.

For example, the common default HTTP port is **8080**.

```
server.port=8081
```

11. Production-Ready Features

Spring Boot Actuator provides endpoints and capabilities useful for observing and managing applications.

- Application health
- Metrics
- Application information
- Monitoring and operational information

```

spring-boot-starter-actuator

/actuator/health

```

12. Advantages of Spring Boot

- Faster application development
- Less manual configuration
- Auto Configuration
- Starter dependencies

- Embedded server
- Easy REST API development
- Easy database integration
- Testing support
- Production-ready monitoring features
- Suitable for microservices and cloud/container deployments

13. Spring Framework vs Spring Boot

Spring Framework	Spring Boot
Core Java application framework	Built on top of Spring Framework
Configuration may require more setup	Reduces setup through auto-configuration
Provides IoC and Dependency Injection	Uses Spring IoC and Dependency Injection
Provides MVC, Data, Security, etc.	Simplifies configuration of these modules
Server setup can be more manual	Supports embedded servers
Dependencies selected individually	Provides starter dependencies
Production setup needs additional work	Provides Actuator and production features

Remember: Spring Boot is not an alternative to Spring Framework. It is built on top of Spring Framework and simplifies Spring application development.

14. Important Interview Points

What is Spring Boot?

A framework built on top of Spring Framework that simplifies the creation of standalone, production-ready Spring applications.

What is Auto Configuration?

Spring Boot automatically configures application components based on dependencies, properties, available beans and configuration conditions.

What is the common default embedded server?

Tomcat for standard Spring MVC web applications.

What are Starter Dependencies?

Predefined dependency groups that simplify dependency management.

What does @SpringBootApplication provide?

Configuration, auto-configuration and component scanning capabilities.

Can Spring Boot be used for microservices?

Yes. It is widely used to build independently deployable REST API and microservice applications.

Does Spring Boot replace Spring Framework?

No. Spring Boot uses Spring Framework internally and simplifies its configuration and usage.